Analytical Chemistry Cumulative Exam
October 19, 2007

1. You are an allergist trying to help a patient who is suffering severe symptoms of an allergy. You test her blood for the antibodies to several allergens and find out that your patient is allergic to cockroach allergen (Bla g 1). In order to help your patient manage her allergy through avoidance, you suggest that she vacuums samples of dust from various places in her apartment and workplace and provide them to you for analysis so that you can see where the sources of Bla g 1 allergen are. You want to determine whether cockroach allergen is above (or below) the asthma morbidity threshold of 8 U per gram of dust in a sample that was vacuumed from the floor in the kitchen. (Note: because the Bla g 1 protein is not well characterized, it is measured in units of "U", which means units, instead of grams. The "U" corresponds to a comparison with activity of a standard solution .) Your laboratory nurse weighs out three, 100-mg aliquots of dust sample, extracts each into a buffer solution and uses an immunoassay to determine the concentrations. Here are the results of those assays: 5.0, 6.6, 7.1 U/g.

(a) (35 points) Can you say at the 95% confidence level if the amount of Bla g 1 in the dust sample is significantly below the asthma morbidity threshold of 8 U / g?

(Show all of your calculations and briefly explain how you arrived at your answer of "yes" or "no".)

(b) (15 points) If the same average and standard deviation that you calculated in part (a) were obtained, but after 10 replicates (instead of 3), would your answer be different than in part (a)? Why? What two parameters have different values in your new calculation? What are their new values?

2. (25 points) Calculate the method precision (overall standard deviation) for the determination of cat allergen (Fel d 1) based on the results from four, 100 mg aliquots from the same dust sample, where analysis in triplicate for each 100 mg gave the following standard deviations: $s_1 = 1.1$ (N=3), $s_2 = 0.5$ (N = 3), $s_3 = 1.4$ (N = 3), $s_4 = 0.5$ (N = 3).

3. (25 points) Calculate the minimum mass of a sample that you should use in an analysis for chloride. Previous analyses of samples from the same geological source have been analyzed to contain about 3.04% chloride. It is known that the method that you will use yields a result for the chloride that is *low* by 0.005 g. In addition, you do not want to exceed a relative error of -0.03 % due to the loss.

Constants, Equations, and Tables that may be of use with the Analytical Cume
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$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

$$E = x_i - x_t$$

$$E_r = \frac{x_i - x_t}{x_t} \times 100\%$$

$$d_i = \left| x_i - \bar{x} \right|$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

$$s = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \bar{x})^2}{N-1}} = \sqrt{\frac{\sum_{i=1}^{N} x_i^2 - \frac{\left(\sum_{i=1}^{N} x_i\right)^2}{N}}{N-1}}$$

$$s_m = \frac{s}{\sqrt{N}}$$

$$s_{pooled} = \sqrt{\frac{\sum_{i=1}^{N_1} (x_i - \overline{x_1})^2 + \sum_{j=1}^{N_2} (x_j - \overline{x_2})^2 + \sum_{k=1}^{N_3} (x_k - \overline{x_3})^2 + \ldots}{N_1 + N_2 + N_3 + \ldots - N_t}}$$

$$s_{pooled} = \sqrt{\frac{(N_1 - 1)\, s_1^2 + (N_2 - 1)\, s_2^2 + (N_3 - 1)\, s_3^2 + \ldots}{N_1 + N_2 + N_3 + \ldots - N_t}}$$

$$y = \frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}} = \frac{e^{-z^2/2}}{\sigma\sqrt{2\pi}}$$

$$z = \frac{(x - \mu)}{\sigma}$$

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t table

Note: t at 12 degrees of freedom and 95% confidence level is 2.19.

Table 7-3

Values of t for Various Levels of Probability

Degrees of freedom	80%	90%	95%	99%	99.9%
1	3.08	6.31	12.7	63.7	637
2	1.89	2.92	4.30	9.92	31.6
3	1.64	2.35	3.18	5.84	12.9
4	1.53	2.13	2.78	4.60	8.61
5	1.48	2.02	2.57	4.03	6.87
6	1.44	1.94	2.45	3.71	5.96
7	1.42	1.90	2.36	3.50	5.41
8	1.40	1.86	2.31	3.36	5.04
9	1.38	1.83	2.26	3.25	4.78
10	1.37	1.81	2.23	3.17	4.59
15	1.34	1.75	2.13	2.95	4.07
20	1.32	1.73	2.09	2.84	3.85
40	1.30	1.68	2.02	2.70	3.55
60	1.30	1.67	2.00	2.62	3.46
∞	1.28	1.64	1.96	2.58	3.29

Confidence Levels for Various Values of z (Table 7.1)

Confidence Level, %	z
50	0.67
68.3	1.00
80	1.28
90	1.64
95	1.96
95.4	2.00
99	2.58
99.7	3.00
99.9	3.29

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